

SUMMARY

EPFL Master of Science specializing in nanotechnology, currently a Research Fellow at NASA JPL developing superconducting single-photon detectors for deep-space communications. Over a year of cleanroom microfabrication experience with research and internship roles at ETH Zurich, EPFL, and Single Quantum, focusing on photonics. Actively seeking research or engineering opportunities in nanotechnology / photonics space.

EDUCATION

EPFL - Swiss Federal Institute of Technology Lausanne, Switzerland	
MSc in Microengineering, Minor in Quantum Science GPA 97% - Top of the class	2023 – 2026
BSc in Microengineering GPA 88%	2019 – 2022
Academic exchanges at TU Delft, Netherlands, and Caltech, California	

EXPERIENCE

NASA Jet Propulsion Laboratory Pasadena, California	Jan 2026 – Present
Graduate Research Fellow	
- Master's thesis on the design and testing of Superconducting Nanowire Single Photon Detectors (SNSPDs), for applications in Deep Space Optical Communications and Photonic Quantum Computing. - Systematically characterized photoresponse and depairing current, improving both the fabrication processes of the superconducting thin films (WSi, NbTiN, MgB₂), and the theoretical understanding of the device physics. - Thermoelectrical simulations and evaluation of state-of-the-art SNSPD photon detection models. - Developed a first of its kind predictive design tool for SNSPDs, which is now submitted for publication.	
Single Quantum Delft, Netherlands	Oct 2025 – Dec 2025
Intern in R&D	
- Led a theoretical and experimental investigation of blackbody radiation induced dark counts in Superconducting Nanowire Single Photon Detectors (SNSPDs) systems. - Developed a physical model and simulation framework to quantify thermal photon generation and propagation through the SNSPD system and its cryostat. This now supports the design of low dark count rate products. - Designed and validated strategies to mitigate dark counts at telecom wavelength, thus improving suitability for applications in Quantum Key Distribution (QKD) and Optical Communications.	
ETH Zurich – PSI Zurich, Switzerland	Jul 2024 – Sep 2024
Summer Intern	
- Designed a compact optical system for double-pass acousto-optic modulation. Achieved a 3x reduction in size with respect to the lab's existing setup, while optimizing for frequency scanning and switching. - Engineered a versatile design for swapping modulators and broaden operating wavelengths (300-900nm). - Deployed the design across setups at ETH Zurich and PSI for optical tweezers and qubit control.	

RESEARCH

Fabrication of Trilayer AFM Cantilevers for Scanning Electrochemical Microscopy Applications.	Feb 2025 – Jul 2025
- Semester thesis at EPFL's LBNI group, on the design and microfabrication of composite (SiN, BCB, Cr, Au, Ti, Pt) atomic force microscopy cantilevers for scanning electrochemical microscopy applications. - Designed and fabricated the devices in cleanroom, experimentally demonstrating novel scanning probe capabilities that resulted in a collaboration with leading industry partners.	
Processes for Niobium Resonators in Circuit Quantum Electrodynamics.	Jul 2024 – Feb 2025
- Semester thesis at EPFL's HQC lab, on the microfabrication and testing of superconducting microwave resonators in Niobium, with a focus on improving internal quality factors. - Identified critical flaws in the thin film processing, providing insights for improved device robustness. - Performed both cleanroom microfabrication and radio frequency characterization in a dilution cryostat.	

SKILLS

Research & Development	Optical Design & Testing, CAD (CATIA, Altium), FEM Simulation (COMSOL), Numerical Modeling, Mechanical Machining, RF/microwave, Data Analysis, Instrumentation Control.
Microfabrication	Process Development, Photolithography, Dry/Wet Etching, Physical Vapor Deposition, Wafer Bonding, Scanning Electron Microscopy, Raman Microscopy, Atomic Force Microscopy.
Software	Python, Git, Matlab, Mathematica, LabView. Intermediate at C++, bash, and html.
Languages	English: <i>Fluent C2</i> , French: <i>Native</i> , Italian: <i>Native</i> , German: <i>Intermediate B2</i> .

CERTIFICATIONS & AWARDS

Best Microengineering MSc Grade Average Prize EPFL	2026
JPL Graduate Fellowship Awardee NASA Jet Propulsion Laboratory	2026
C2 Certificate in Advanced English Cambridge University	2019
Prize for Excellence in Mathematics and Sciences Nestlé	2016

OTHER ACTIVITIES

Teaching	Tutoring students up to graduate level (Freelance, University TA, and in Private school).
Hobbies	Volleyball player, Endurance runner, Guitarist, and Mathematics hobbyist.